

Homework Classical Physics, Fall 2023

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Problem Set 01.2 (L01 & L02)

Due on Thursday, 14. May 2023, at 09:30 (in the morning!)
(= Deadline for submitting a pdf file with solution)

Mechanics - derivatives and integrals (4 points)

Calculate the derivative of the functions with respect to x or t

a) $f(x) = \frac{1}{4}x^2 - \frac{3}{4}\frac{1}{x^2}$

b) $g(t) = a \ln(3bt^3)$

c) $f(x) = 5(x+1)^2 \cos(x)$

Calculate the indefinite integrals of the functions with respect to x or t

d) $f(x) = 3e^{ax}$

e) $f(x) = \frac{1}{4}x^2 - \frac{3}{4}\frac{1}{x^2}$

f) $g(t) = \frac{3}{4}b^2 \cos(t)$

Don't know the rules - check out the list of derivatives and integrals in Moodle under Official Course Files